

Predicting EMG Envelopes from EEG and Kinematic Signals:

A Research Article Proposal

MohammadMahdi SharifBeigi

Computational Neuroscience Team 6

Abstract

Electromyography (EMG)-based prosthetic control fails in cases of high-level amputation or severe paralysis where functional muscles are absent. Brain-derived signals from electroencephalography (EEG) offer a complementary pathway. This article presents a **multi-method comparative study** for continuous prediction of EMG envelopes from scalp EEG and kinematic measurements during grasp-and-lift tasks, validated on the **WAY-EEG-GAL** dataset (3,936 trials, 12 participants, 32-channel EEG, 5-channel EMG, 3D kinematics, 6-axis force/torque). We propose and compare four architectures of increasing complexity:

Method 1 (KG-GT): A Kinematic-Guided Graph-Transformer combining CCA-based EEG–kinematic alignment, a Transformer encoder, and a kinematic-conditioned GAT with a hybrid MSE + SoftDTW loss.

Method 2 (NNMF-KG-GT): Enhances Method 1 by replacing the anatomically fixed graph structure with Non-Negative Matrix Factorisation (NNMF)-derived muscle synergy weights, providing biologically interpretable, data-driven inter-muscle connectivity.

Method 3 (PTL-Net): A Phase-Targeted Lightweight Network that focuses on Load and Hold movement phases, uses correlation-based channel selection (6 channels: C3, C4, Cz, CP3, CP4, Fz), and performs multi-task prediction of EMG envelopes, Grip Force, and object weight classification—demonstrating direct clinical applicability for adaptive prosthetic control.

Method 4 (USPT): A Unified Synergy-Phase-Transformer that integrates all innovations from Methods 1–3 into a single framework with NNMF-initialised kinematic-guided GAT, phase-aware Transformer encoding, triple-head decoding, and phase-weighted loss.

Keywords: EEG, EMG prediction, WAY-EEG-GAL, NNMF, muscle synergy, graph attention network, Transformer, phase-targeted decoding, multi-task learning, prosthetic control.

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1 Introduction

The ability to decode motor intent from neural signals and translate it into reliable control of external devices is a foundational challenge in neural engineering. Two principal peripheral modalities have historically dominated prosthetic control: surface electromyography (sEMG), which measures the electrical activity of muscles, and joint kinematics, which captures the positional state of limbs. While both are effective for able-bodied individuals and low-level amputees, their availability becomes severely restricted in cases of high-level limb loss, spinal cord injury, or progressive neuromuscular disease.

Electroencephalography (EEG) offers a non-invasive, low-cost alternative by capturing oscillatory cortical activity through scalp-mounted electrodes. Motor-related cortical areas exhibit characteristic modulations in the mu (8–13 Hz) and beta (13–30 Hz) frequency bands during movement preparation and execution, as well as slow cortical potentials in the delta band (0.1–3 Hz) that track the continuous dynamics of movement.

The core scientific question addressed here is: *to what extent can the continuous EMG envelope be accurately predicted from scalp EEG recordings, and can the simultaneous inclusion of kinematic measurements, muscle synergy decomposition, and phase-targeted processing further improve this prediction?*

This work makes the following contributions:

- A novel **CCA-based EEG–kinematic alignment** stage (Method 1) and a **Corticomuscular Coherence (CMC)-based EEG–EMG connectivity** feature extraction (Method 4) that provide a biologically-grounded projection before deep learning.
- A **time-lagged windowing strategy** to account for physiological delays between cortical planning, muscle activation, and limb kinematics [13].
- A **Transformer encoder** that explicitly models the nonlinear, multi-lag temporal coupling between cortical activity and muscle activation (Methods 1, 2, 4).
- A **Kinematic-Guided GAT** whose attention coefficients are dynamically conditioned on the instantaneous kinematic state, enabling task-phase-dependent muscle synergy modelling (Methods 1, 2, 4).
- Integration of **NNMF-derived muscle synergy** weights as data-driven graph adjacency priors, inspired by multimodal approaches to motor dynamics [9] (Methods 2, 4).
- A **phase-targeted, channel-selected, multi-task** architecture that predicts EMG envelopes, Grip Force, and object weight class simultaneously from only 6 EEG

channels during Load and Hold phases (Methods 3, 4).

- A **Unified Synergy-Phase-Transformer (USPT)** combining all innovations with phase-weighted loss and staged training (Method 4).
- A **hybrid MSE + SoftDTW loss** function that penalises both amplitude error and temporal misalignment (all methods).
- Comprehensive validation on the richly multimodal **WAY-EEG-GAL** dataset across 12 subjects and systematically varied grasp conditions.

2 Background and Related Work

2.1 The WAY-EEG-GAL Dataset

The WAY-EEG-GAL dataset [1] contains 3,936 grasp-and-lift (GAL) trials from 12 right-handed participants. The participant grasped a small instrumented object with the thumb and index finger, lifted it approximately 5 cm, held it at a target position, and replaced it. Object weight was varied between 165, 330, and 660 g; contact surface was varied between sandpaper, suede, and silk—in an unpredictable manner—enforcing systematic changes in fingertip force coordination. Simultaneous recordings include:

- **EEG:** 32 channels (ActiCap), sampled at 500 Hz.
- **EMG:** 5 channels (anterior deltoid, brachioradialis, flexor digitorum, extensor digitorum, first dorsal interosseus), sampled at 4,000 Hz.
- **Kinematics:** Four Polhemus FASTRAK 3D position/orientation sensors (object, index finger, thumb, wrist), sampled at 500 Hz.
- **Force/torque:** Two ATI Nano-17 6-axis sensors at both contact plates.

Sixteen behaviourally relevant events were extracted per trial, and all participants were confirmed to adapt grip coordination to the prevailing weight and friction [1].

2.2 Unscented Kalman Filtering Approaches

Nakagome et al. [2] demonstrated offline continuous decoding of lower-limb EMG envelopes during over-ground walking on multiple terrains from delta-band (0.1–3 Hz) EEG using a first-order Unscented Kalman Filter (UKF). Across six subjects the mean Pearson correlation was $r = 0.236$, with 79.3% of results exceeding $\text{SNR} = 0$ dB.

Sburlea, Butturini, and Müller-Putz [3] extended the UKF approach to upper-limb grasping (31 subjects, 33 grasping conditions) using low-frequency EEG (0.1–2 Hz), a multivari-

ate autoregressive (MVAR) state model, and partial least squares (PLS) neural tuning, achieving median $r = 0.20$ and nMAE = 28.6%.

The UKF state-transition equation is:

$$\mathbf{x}_n = \sum_{i=1}^m \mathbf{A}_i(n) \mathbf{x}_{n-i} + \mathbf{e}_n, \quad (1)$$

where $\mathbf{x}_n \in \mathbb{R}^d$ is the EMG envelope state, $\mathbf{A}_i(n)$ are $d \times d$ coefficient matrices, and $\mathbf{e}_n$ is Gaussian noise. The observation model is:

$$\mathbf{y}_k = \mathbf{H} \mathbf{x}_k + \mathbf{v}_{k-1}, \quad (2)$$

where $\mathbf{H}$ is the PLS-estimated neural tuning matrix and $\mathbf{v}$ is measurement noise. The UKF propagates sigma points through Eq. (1) to estimate state statistics nonlinearly. However, the *fixed linear* observation model $\mathbf{H}$ and the Gaussian noise assumptions limit performance ceiling for complex grasping.

2.3 Deep Sequence Learning: CNN-LSTM

Jain et al. [4] proposed *RawSeq2SeqNet*, a CNN-LSTM architecture for reconstructing biceps brachii EMG from 19-channel EEG during elbow flexion-extension (single subject). The key innovation was a hybrid training loss:

$$\mathcal{L} = 0.9 \cdot \text{MSE} + 0.1 \cdot \text{SoftDTW}(\gamma = 0.2), \quad (3)$$

achieving $R^2 = 0.364$ and Pearson $r = 0.795$ on the held-out test set, with mean inference time 2.11 ms per 2-second window. The SoftDTW component tolerates the physiological cortico-muscular conduction delay ($\approx 10\text{--}30$ ms) during training.

2.4 Graph-Based Architectures with Muscle Synergy

Li et al. [5] introduced *MSGAT-LSTM* for continuous wrist kinematics prediction from sparse sEMG, using a GAT with edge weights derived from cosine similarity of EMG feature vectors, grounded in muscle synergy (MS) theory. The cosine similarity between nodes i and j after shared linear transform $\mathbf{W}$ is:

$$\alpha_{ij} = \frac{\sum_{k=1}^{F'} m_{ik} m_{jk}}{\|\mathbf{M}_i\| \|\mathbf{M}_j\|}, \quad (4)$$

with normalised edge weights:

$$e_{ij} = \frac{\exp(\text{LeakyReLU}(\alpha_{ij}))}{\sum_{j'} \exp(\text{LeakyReLU}(\alpha_{ij'}))}. \quad (5)$$

On NinaPro DB2 (12 subjects), MSGAT-LSTM achieved $R^2 = 0.960$ and RMSE = 0.098 rad using only 10 sparse electrodes, reducing training time by $\approx 13\%$ relative to GCN-LSTM.

2.5 Non-Negative Matrix Factorisation for Muscle Synergy

Pierella et al. [9] demonstrated a multimodal approach to capturing post-stroke motor recovery dynamics by integrating EEG, EMG, and kinematic data. A central analytical tool was **Non-Negative Matrix Factorisation (NNMF)** [10], which decomposes the multi-muscle EMG activation matrix $\mathbf{V} \in \mathbb{R}_{\geq 0}^{T \times M}$ into a small number r of synergy modules:

$$\mathbf{V} \approx \mathbf{W} \mathbf{H}, \quad \mathbf{W} \in \mathbb{R}_{\geq 0}^{T \times r}, \quad \mathbf{H} \in \mathbb{R}_{\geq 0}^{r \times M}, \quad (6)$$

where $\mathbf{W}$ encodes temporal activation profiles and $\mathbf{H}$ encodes muscle recruitment weights for each synergy. The non-negativity constraint ensures physiological interpretability: muscles can only be activated, not inhibited, by a synergy module.

Pierella et al. showed that the number of synergies, their composition, and their temporal activation patterns change during post-stroke rehabilitation, tracking recovery of motor coordination at the spinal level. Tresch et al. [11] provided foundational evidence that the central nervous system controls movement through a small set of muscle synergies, validating NNMF as a principled decomposition for motor neuroscience.

Gap. Despite the established utility of NNMF for synergy analysis, no prior EEG-to-EMG prediction study has used NNMF-derived synergy weights as structural priors for graph-based neural network architectures. This gap is addressed by Methods 2 and 4 in the present work.

3 Research Gaps and Motivation

The literature review reveals eight specific gaps:

G1. Limited expressivity. UKF imposes linear, Gaussian assumptions on the EEG–EMG mapping; deep architectures with transformer-level capacity are needed.

G2. Single-subject scope. RawSeq2SeqNet was validated on one participant; the WAY-EEG-GAL dataset provides 12 participants and 3,936 trials.

- G3. Underutilisation of kinematics.** No existing EEG-to-EMG predictor leverages the rich kinematic data (3D finger positions, grip aperture, load/grip force) available in WAY-EEG-GAL as a structural constraint.
- G4. No synergy structure in EEG decoding.** The muscle-synergy graph principle proven in MSGAT-LSTM has never been applied to the EEG-to-EMG direction.
- G5. Temporal misalignment.** With the exception of Jain et al., prior EEG-to-EMG methods use MSE/MAE loss functions sensitive to cortico-muscular conduction delays.
- G6. No NNMF-derived priors for graph construction.** Existing graph-based architectures use either anatomically fixed or learned-from-scratch graph topologies. NNMF-derived synergy weights—validated by decades of motor neuroscience research [9, 11]—have never been used to initialise the adjacency matrix of a GAT for EEG-to-EMG prediction.
- G7. Full-window prediction ignores phase-specific coupling.** All prior methods predict EMG uniformly across the entire trial window. However, the EEG–EMG relationship is strongest during Load and Hold phases, where sustained cortical drive maintains isometric contraction. Reach and Replace phases have weaker coupling and higher artefact contamination.
- G8. No multi-task or classification from EEG.** No EEG-to-EMG study simultaneously predicts EMG envelopes, grip force, and object weight class. Multi-task learning can improve representation quality through shared features, and weight classification directly demonstrates clinical applicability for adaptive prosthetic grip control.

4 Methods Overview

This work proposes four methods of increasing complexity, each addressing a progressively larger subset of the identified research gaps. Table 1 summarises the four methods.

Table 1: Overview of the four proposed methods.

Method	Name	Key Innovation	Gaps Addressed
1	KG-GT	CCA + Transformer + Kinematic-Guided GAT + SoftDTW	G1–G5
2	NNMF-KG-GT	Method 1 + NNMF-derived synergy graph adjacency	G1–G6
3	PTL-Net	Phase targeting + channel selection + multi-task (EMG + Force + weight class)	G7–G8
4	USPT	Unified: NNMF + phase embeddings + channel selection + CMC + KG-GAT + triple-head	G1–G9

5 Method 1: Kinematic-Guided Graph-Transformer (KG-GT)

5.1 Method Overview

The KG-GT architecture addresses gaps G1–G5 by integrating three core innovations: (1) CCA-based EEG–kinematic alignment for biomechanically grounded feature projection; (2) a Transformer encoder for modelling nonlinear, multi-lag cortico-muscular temporal coupling; and (3) a Kinematic-Guided Graph Attention Network encoding dynamic, task-phase-dependent muscle synergy relationships. Training employs a hybrid MSE + SoftDTW loss to penalise both amplitude error and temporal misalignment.

5.1.1 Gap Addressed and Solution

Table 2: Method 1 gap–solution mapping.

Gap	Solution in Method 1
G1: Limited expressivity of UKF	Transformer encoder replaces linear state-space model
G2: Single-subject scope	Validated on all 12 WAY-EEG-GAL subjects (3,936 trials)
G3: Underutilisation of kinematics	CCA alignment + kinematic conditioning of GAT
G4: No synergy structure in EEG decoding	GAT over 5 EMG channels with dynamic edge weights
G5: Temporal misalignment	Hybrid MSE + SoftDTW loss

5.2 Preprocessing Pipeline

5.2.1 EEG Preprocessing

Raw 32-channel EEG (500 Hz) is processed as follows:

1. Fourth-order zero-phase Butterworth bandpass filter: 0.1–40 Hz.
2. 50 Hz notch filter (power-line).
3. Artifact Subspace Reconstruction (ASR): sliding window 500 ms, threshold 5 std from a noise-free baseline.
4. Common Average Reference (CAR) re-referencing.
5. Delta-band extraction (0.1–2 Hz) via fourth-order zero-phase Butterworth.
6. Windowing: $T_w = 500$ samples (1 s) with stride 50 samples (100 ms), yielding $\mathbf{X}_{\text{raw}} \in \mathbb{R}^{N \times 500 \times 32}$.

5.2.2 EMG Preprocessing

Raw 5-channel EMG (4,000 Hz) envelope extraction:

1. Bandpass filter 30–300 Hz (4th-order Butterworth).
2. Full-wave rectification: $|s(t)|$.
3. Low-pass filter at 10 Hz to obtain smooth activation envelope.

4. Downsample to 500 Hz; z-score normalise per channel over training set.
5. Target tensor: $\mathbf{Y} \in \mathbb{R}^{N \times 500 \times 5}$.

5.2.3 Kinematic Preprocessing

Position signals are differentiated via Savitzky-Golay filtering. The kinematic state vector at each time step is:

$$\mathbf{k}_t = [\mathbf{p}_{\text{wrist}}, \mathbf{p}_{\text{index}}, \mathbf{p}_{\text{thumb}}, d_{\text{grip}}, F_L, F_G, \rho_{GL}]^\top \in \mathbb{R}^{13}, \quad (7)$$

where $\mathbf{p}$ are 3D positions, d_{grip} is grip aperture (Euclidean distance between index and thumb sensors), F_L and F_G are load and grip forces, and $\rho_{GL} = F_G/F_L$ is their ratio. Kinematic tensor: $\mathbf{K} \in \mathbb{R}^{N \times 500 \times 13}$.

5.2.4 Method 1 Preprocessing Pipeline Diagram

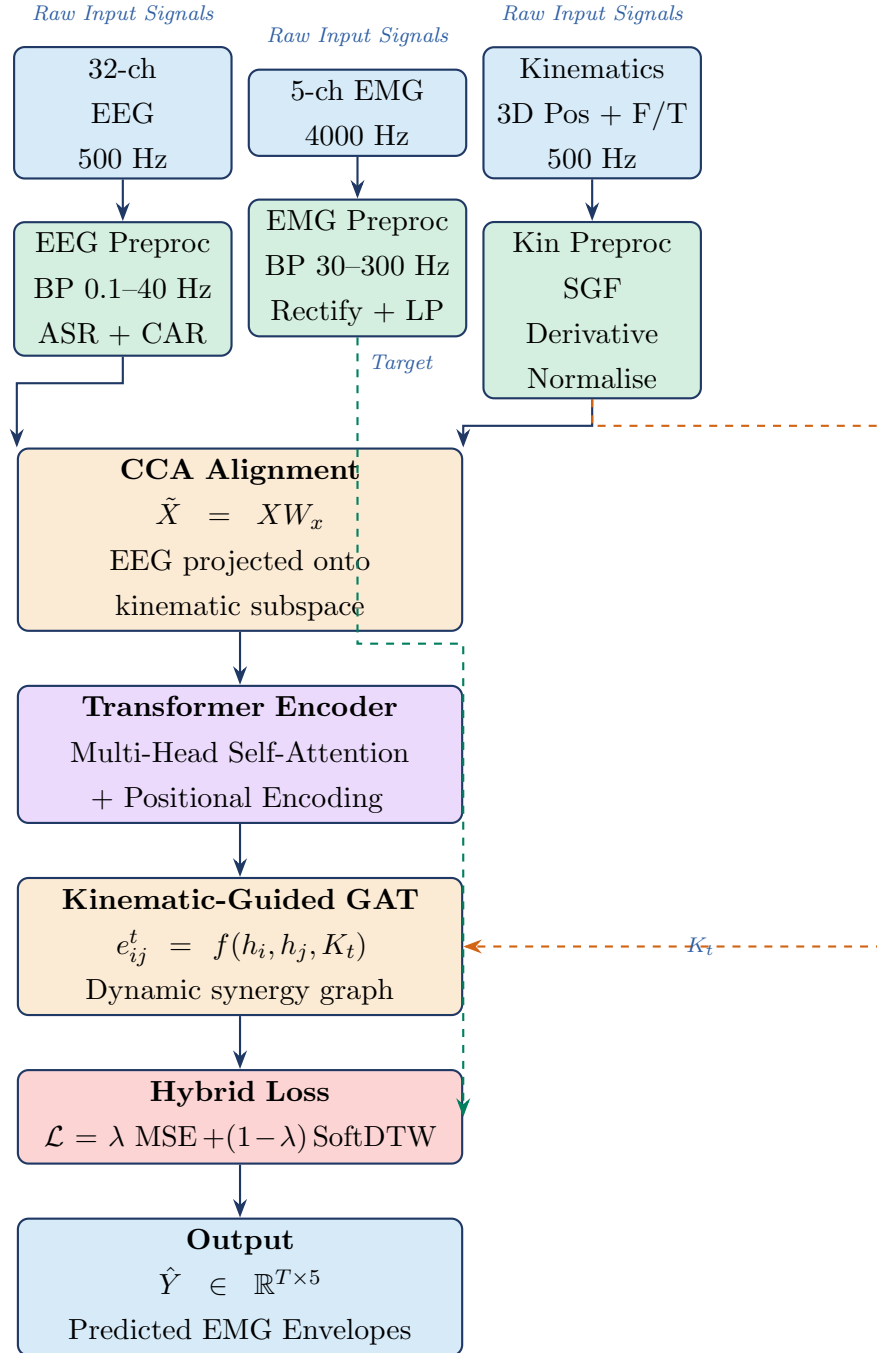


Figure 1: End-to-end processing pipeline of Method 1 (KG-GT). EEG and kinematic data are jointly aligned via CCA before entering the Transformer encoder and the kinematic-guided GAT. The dashed green arrow shows the EMG envelope used as the training target; the dashed orange arrow shows kinematic state injected into the GAT at inference time.

5.3 Model Architecture

5.3.1 CCA-Based EEG–Kinematic Feature Alignment

Canonical Correlation Analysis (CCA) aligns the EEG feature space with the kinematic measurements by finding projection matrices $\mathbf{W}_x \in \mathbb{R}^{32 \times d}$ and $\mathbf{W}_k \in \mathbb{R}^{13 \times d}$ that solve:

$$\max_{\mathbf{W}_x, \mathbf{W}_k} \rho = \frac{\mathbf{W}_x^\top \Sigma_{xk} \mathbf{W}_k}{\sqrt{\mathbf{W}_x^\top \Sigma_{xx} \mathbf{W}_x} \sqrt{\mathbf{W}_k^\top \Sigma_{kk} \mathbf{W}_k}}, \quad (8)$$

where Σ_{xx} , Σ_{kk} are within-set covariances and Σ_{xk} is the cross-covariance matrix, estimated from the training set and solved via generalised eigenvalue decomposition. The EEG data are then projected:

$$\tilde{\mathbf{X}} = \mathbf{X} \mathbf{W}_x, \quad \tilde{\mathbf{X}} \in \mathbb{R}^{T \times d}, \quad d = 16. \quad (9)$$

This reduces dimensionality from 32 channels to d canonical components while encoding biomechanical prior knowledge into the feature representation.

Figure 2 illustrates the CCA alignment stage.

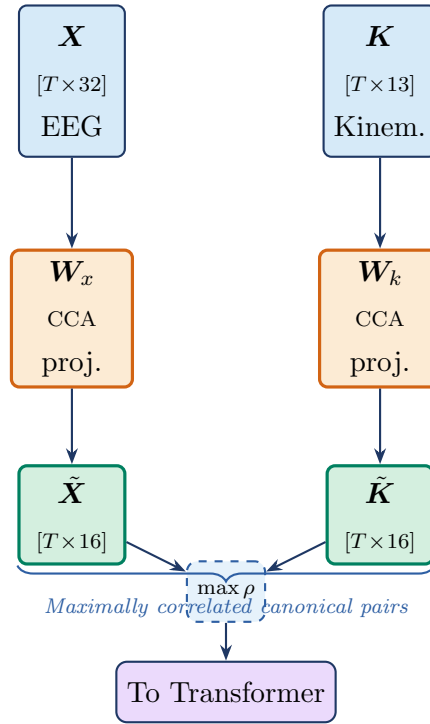


Figure 2: CCA alignment stage. EEG matrix $\mathbf{X}$ and kinematic matrix $\mathbf{K}$ are projected via learned matrices $\mathbf{W}_x$ and $\mathbf{W}_k$ that maximise the canonical correlation ρ . Only $\tilde{\mathbf{X}}$ propagates to the Transformer; $\tilde{\mathbf{K}}$ serves as the alignment target during training.

5.3.2 Transformer Encoder for Temporal Modelling

Sinusoidal positional encoding is added to the CCA-projected input:

$$\text{PE}_{(p,2i)} = \sin\left(\frac{p}{10000^{2i/d}}\right), \quad (10)$$

$$\text{PE}_{(p,2i+1)} = \cos\left(\frac{p}{10000^{2i/d}}\right), \quad (11)$$

$$\mathbf{Z}^{(0)} = \tilde{\mathbf{X}} + \mathbf{P}\mathbf{E}. \quad (12)$$

Each of the L stacked Transformer layers applies Multi-Head Self-Attention (MHSA) followed by a Feed-Forward Network (FFN) with residual connections:

$$\mathbf{Q}_i = \mathbf{Z}^{(l-1)}\mathbf{W}_{Q_i}, \quad \mathbf{K}_i = \mathbf{Z}^{(l-1)}\mathbf{W}_{K_i}, \quad \mathbf{V}_i = \mathbf{Z}^{(l-1)}\mathbf{W}_{V_i}, \quad (13)$$

$$\text{head}_i = \text{softmax}\left(\frac{\mathbf{Q}_i\mathbf{K}_i^\top}{\sqrt{d_k}}\right)\mathbf{V}_i, \quad (14)$$

$$\text{MHSA}(\mathbf{Z}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)\mathbf{W}_O, \quad (15)$$

$$\text{FFN}(\mathbf{x}) = \max(0, \mathbf{x}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2. \quad (16)$$

The full layer update with layer normalisation (LN) and residual connections is:

$$\mathbf{Z}'^{(l)} = \text{LN}(\mathbf{Z}^{(l-1)} + \text{MHSA}(\mathbf{Z}^{(l-1)})), \quad (17)$$

$$\mathbf{Z}^{(l)} = \text{LN}(\mathbf{Z}'^{(l)} + \text{FFN}(\mathbf{Z}'^{(l)})). \quad (18)$$

The output $\mathbf{H}_{\text{temp}} = \mathbf{Z}^{(L)} \in \mathbb{R}^{T \times d_{\text{model}}}$ is a temporally contextualised encoding accounting for multi-lag dependencies within the physiological delay range (5–25 samples at 500 Hz).

Figure 3 illustrates one Transformer layer.

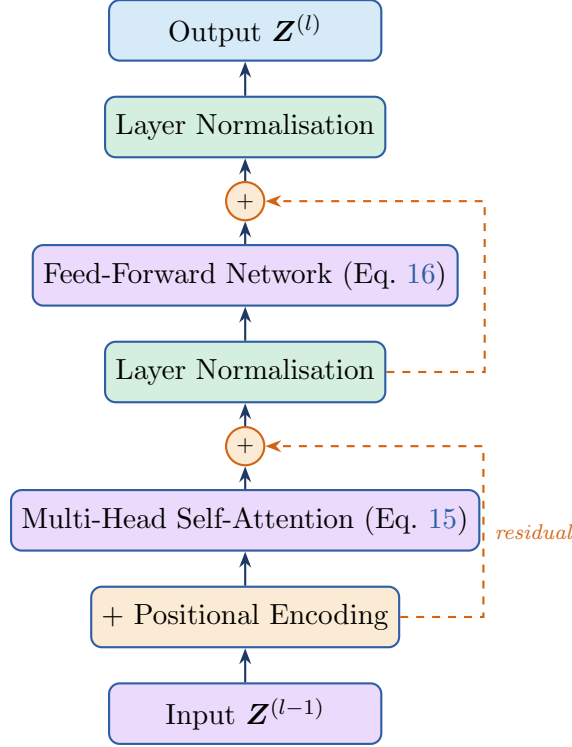


Figure 3: One Transformer encoder layer. Residual (skip) connections (dashed orange) prevent vanishing gradients and preserve low-level temporal features across depth.

5.3.3 Kinematic-Guided Graph Attention Network

A graph $G = (V, E)$ is defined where each node $v_i \in V$ represents one of the $M = 5$ EMG channels and edges E encode potential muscle synergy. The key innovation is that attention coefficients are dynamically conditioned on the kinematic state vector $\mathbf{k}_t$ (Eq. 7):

$$e_{ij}^t = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j \parallel \mathbf{W}_k\mathbf{k}_t]), \quad (19)$$

where $\mathbf{h}_i$ is the feature vector of node i extracted from $\mathbf{H}_{\text{temp}}$, $\parallel$ denotes concatenation, $\mathbf{W}$ and $\mathbf{W}_k$ are learnable matrices, and $\mathbf{a}$ is a learnable attention vector. Normalised:

$$\alpha_{ij}^t = \frac{\exp(e_{ij}^t)}{\sum_{n \in \mathcal{N}_i} \exp(e_{in}^t)}. \quad (20)$$

Node features are updated as:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij}^t \mathbf{W}\mathbf{h}_j^{(l)} \right), \quad (21)$$

and the Jumping Knowledge (JK) strategy aggregates representations from both graph layers:

$$\mathbf{h}_i^{\text{final}} = \text{Concat}(\mathbf{h}_i^{(1)}, \mathbf{h}_i^{(2)}). \quad (22)$$

The decoder maps to the predicted EMG envelope:

$$\hat{\mathbf{Y}} = \mathbf{H}_{\text{final}} \mathbf{W}_{\text{out}} + \mathbf{b}_{\text{out}}, \quad \hat{\mathbf{Y}} \in \mathbb{R}^{T \times M}. \quad (23)$$

Figure 4 illustrates the kinematic-guided GAT over the five muscle nodes.

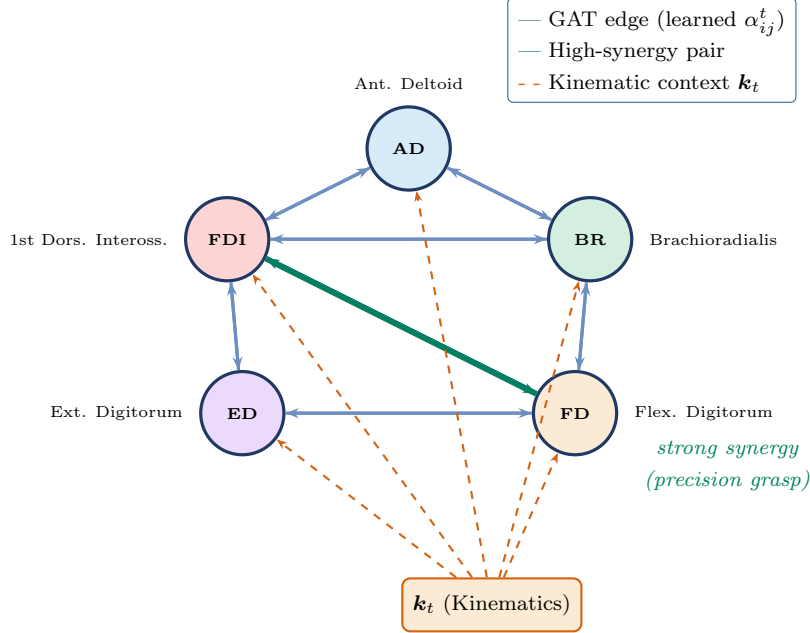


Figure 4: Kinematic-Guided GAT over the five WAY-EEG-GAL EMG channels. AD = Anterior Deltoid, BR = Brachioradialis, FD = Flexor Digitorum, ED = Extensor Digitorum, FDI = First Dorsal Interosseus. Edge weights α_{ij}^t are dynamically conditioned on the kinematic state $\mathbf{k}_t$, enabling task-phase-dependent synergy. The thick green edges indicate the expected strong synergy between FD and FDI during the precision grasp loading phase.

5.3.4 Time-Aware Hybrid Loss Function

Training minimises:

$$\mathcal{L}_{\text{total}} = \lambda \text{MSE}(\mathbf{Y}, \hat{\mathbf{Y}}) + (1 - \lambda) \text{SoftDTW}_{\gamma}(\mathbf{Y}, \hat{\mathbf{Y}}), \quad (24)$$

where:

$$\text{MSE}(\mathbf{Y}, \hat{\mathbf{Y}}) = \frac{1}{TM} \sum_{t=1}^T \sum_{m=1}^M (y_{t,m} - \hat{y}_{t,m})^2, \quad (25)$$

$$\text{SoftDTW}_{\gamma}(\mathbf{Y}, \hat{\mathbf{Y}}) = \min_{\pi}^{\gamma} \langle \mathbf{D}(\mathbf{Y}, \hat{\mathbf{Y}}), \mathbf{A}(\pi) \rangle, \quad (26)$$

and the soft-minimum operator [6] is:

$$\min^{\gamma}(a_1, \dots, a_n) = -\gamma \log \sum_{i=1}^n \exp\left(-\frac{a_i}{\gamma}\right). \quad (27)$$

As $\gamma \rightarrow 0$ this reduces to standard (non-differentiable) DTW; a value $\gamma = 0.1$ balances gradient smoothness and alignment precision. The initial value $\lambda = 0.9$ follows Jain et al. [4] and will be tuned on the validation set.

5.3.5 Complete Architecture Diagram

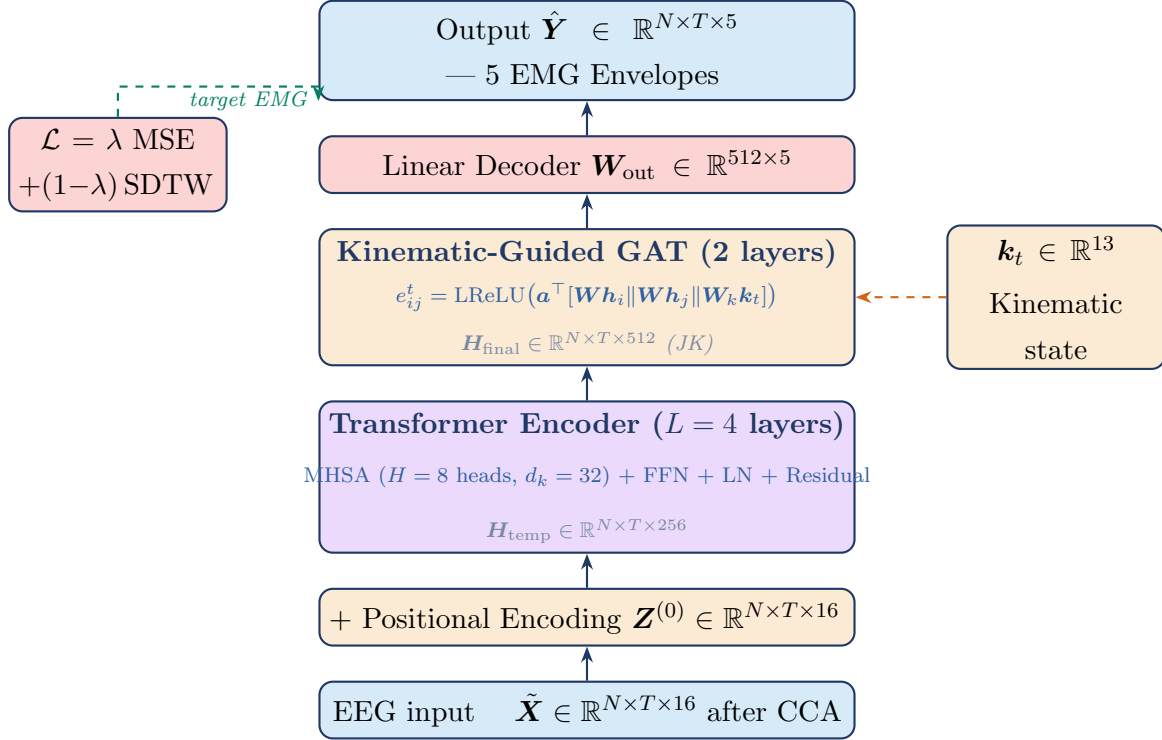


Figure 5: Complete KG-GT architecture (Method 1). Numbers in italics indicate tensor dimensions at each stage. The dashed orange arrow shows kinematic state injected into the GAT; the dashed green arrow shows the EMG training target fed to the hybrid loss.

5.4 Training Protocol

The model is trained with Adam ($\eta = 10^{-3}$, halved every 50 epochs without validation improvement), batch size 32, early stopping (patience 30 epochs), dropout $p = 0.2$, and gradient clipping at norm 1.0. Data are split at the series level: 7 series per participant for training, 1 for validation, 2 (held-out weight and mixed series) for testing. Leave-one-subject-out cross-validation (LOSOCV) is also reported.

5.5 Evaluation Metrics

Method 1 employs the following evaluation metrics (shared with all subsequent methods unless otherwise noted):

Normalised RMSE

$$\text{nRMSE}_m = \frac{1}{R_m} \sqrt{\frac{1}{T} \sum_{t=1}^T (y_{t,m} - \hat{y}_{t,m})^2}. \quad (28)$$

Normalised MAE

$$\text{nMAE}_m = \frac{1}{T R_m} \sum_{t=1}^T |y_{t,m} - \hat{y}_{t,m}|. \quad (29)$$

Pearson Correlation

$$r_m = \frac{\text{Cov}(Y_m, \hat{Y}_m)}{\sqrt{\text{Var}(Y_m) \text{Var}(\hat{Y}_m)}}. \quad (30)$$

Coefficient of Determination

$$R_m^2 = 1 - \frac{\sum_t (y_{t,m} - \hat{y}_{t,m})^2}{\sum_t (y_{t,m} - \bar{y}_m)^2}. \quad (31)$$

Signal-to-Noise Ratio

$$\text{SNR}_m = 10 \log_{10} \left[\frac{\text{Var}(Y_m)}{\text{MSE}(Y_m, \hat{Y}_m)} \right]. \quad (32)$$

6 Method 2: NNMF-Enhanced KG-GT

6.1 Method Overview

Method 2 enhances the KG-GT architecture by replacing the anatomically fixed graph structure with a **data-driven synergy graph** derived from Non-Negative Matrix Factorisation (NNMF) [10]. Inspired by Pierella et al. [9], who demonstrated that NNMF can decompose multi-muscle EMG recordings into a small number of synergy modules capturing the temporal dynamics of motor coordination, we apply NNMF to the WAY-EEG-GAL EMG training data to extract biologically interpretable muscle synergy weights. These weights then serve as the adjacency matrix for the GAT, providing a principled, data-driven encoding of inter-muscle coupling that adapts to the specific coordination patterns of each participant.

6.1.1 Gap Addressed and Solution

Table 3: Method 2 gap-solution mapping.

Gap	Solution in Method 2
G4: No synergy structure in EEG decoding	NNMF-derived synergy weights define the GAT adjacency matrix
G6: No NNMF-derived priors for graph construction	Cosine similarity of NNMF synergy loadings replaces fixed anatomy
G1–G3, G5: (inherited from Method 1)	Transformer + kinematic conditioning + Soft-DTW loss retained

6.2 Preprocessing Pipeline

6.2.1 EEG and Kinematic Preprocessing

Identical to Method 1 (Section 5): 32-channel EEG undergoes bandpass filtering (0.1–40 Hz), notch at 50 Hz, ASR artifact removal, CAR re-referencing, and delta-band extraction. Kinematic state vectors $\mathbf{k}_t \in \mathbb{R}^{13}$ are constructed with Savitzky-Golay differentiation. CCA alignment (Eq. 8–9) is applied to project EEG onto the kinematic subspace.

6.2.2 EMG Preprocessing and NNMF Decomposition

The standard EMG envelope extraction from Method 1 is performed first (bandpass 30–300 Hz, rectification, low-pass at 10 Hz, downsampling to 500 Hz). The critical addition is

the NMF decomposition applied to the training-set EMG envelopes.

Non-Negative Matrix Factorisation. Given the concatenated EMG envelope matrix $\mathbf{V} \in \mathbb{R}_{\geq 0}^{T_{\text{total}} \times M}$ (where T_{total} is the total number of training time steps and $M = 5$ muscles), NMF seeks two non-negative factor matrices:

$$\mathbf{V} \approx \mathbf{W} \mathbf{H}, \quad \mathbf{W} \in \mathbb{R}_{\geq 0}^{T_{\text{total}} \times r}, \quad \mathbf{H} \in \mathbb{R}_{\geq 0}^{r \times M}, \quad (33)$$

where r is the number of synergy modules and $\mathbf{H}$ is the **synergy weight matrix** whose rows encode how each synergy recruits the five muscles. The factorisation minimises the Frobenius norm:

$$\min_{\mathbf{W} \geq 0, \mathbf{H} \geq 0} \|\mathbf{V} - \mathbf{W} \mathbf{H}\|_F^2 = \sum_{t,m} (V_{tm} - (\mathbf{W} \mathbf{H})_{tm})^2. \quad (34)$$

Multiplicative Update Rules [10]. The factors are iteratively updated as:

$$H_{rm} \leftarrow H_{rm} \frac{(\mathbf{W}^\top \mathbf{V})_{rm}}{(\mathbf{W}^\top \mathbf{W} \mathbf{H})_{rm}}, \quad (35)$$

$$W_{tr} \leftarrow W_{tr} \frac{(\mathbf{V} \mathbf{H}^\top)_{tr}}{(\mathbf{W} \mathbf{H} \mathbf{H}^\top)_{tr}}, \quad (36)$$

which guarantee monotonic decrease of the objective and preservation of non-negativity.

Selection of Synergy Number r . The optimal r is determined by the Variance Accounted For (VAF):

$$\text{VAF} = 1 - \frac{\|\mathbf{V} - \mathbf{W} \mathbf{H}\|_F^2}{\|\mathbf{V}\|_F^2}. \quad (37)$$

We select the smallest r such that $\text{VAF} \geq 0.90$, consistent with the threshold used by Pierella et al. [9] and Tresch et al. [11]. For grasp-and-lift tasks involving five muscles, we expect $r \in \{2, 3\}$.

6.2.3 NNMF Decomposition Diagram

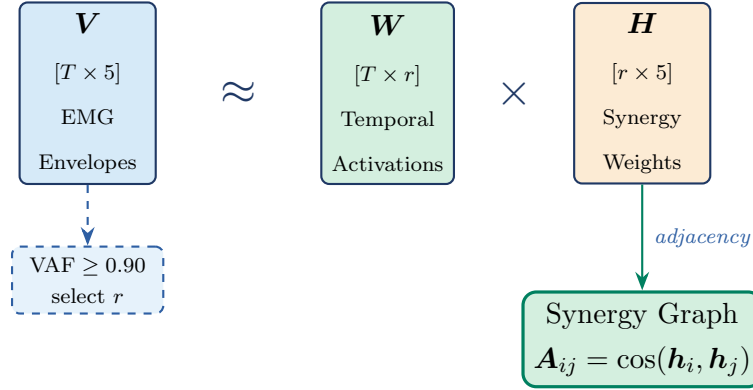


Figure 6: NNMF decomposition of the EMG envelope matrix $\mathbf{V}$ into temporal activation patterns $\mathbf{W}$ and synergy weight matrix $\mathbf{H}$. The rows of $\mathbf{H}$ define muscle recruitment patterns; their pairwise cosine similarity constructs the synergy adjacency matrix $\mathbf{A}$.

6.3 Model Architecture

6.3.1 NNMF-Based Synergy Graph Construction

The synergy weight matrix $\mathbf{H} \in \mathbb{R}_{\geq 0}^{r \times M}$ encodes how each of the r synergy modules recruits the $M = 5$ muscles. The column vector $\mathbf{h}_m = \mathbf{H}_{:,m} \in \mathbb{R}^r$ captures the synergy “fingerprint” of muscle m . The adjacency matrix of the synergy graph is defined as:

$$A_{ij} = \frac{\mathbf{h}_i^\top \mathbf{h}_j}{\|\mathbf{h}_i\| \|\mathbf{h}_j\|}, \quad i, j \in \{1, \dots, M\}, \quad (38)$$

i.e., the cosine similarity of the synergy loading vectors. Muscles that are co-recruited by the same synergies receive high adjacency weights, providing a biologically grounded prior for the graph structure.

Comparison with Method 1. In Method 1, the graph adjacency was defined anatomically (fully connected with uniform initialisation) and learned implicitly through GAT attention. In Method 2, $\mathbf{A}$ is *pre-computed* from training-set NNMF and used to *initialise* the GAT edge mask, biasing the network towards physiologically meaningful synergy patterns from the outset.

6.3.2 Synergy Similarity Index

To validate the extracted synergies against known motor control literature, we compute the Synergy Similarity Index (SSI) between synergies extracted from different subjects:

$$\text{SSI}(\mathbf{h}_i^{(a)}, \mathbf{h}_i^{(b)}) = \frac{\mathbf{h}_i^{(a)\top} \mathbf{h}_i^{(b)}}{\|\mathbf{h}_i^{(a)}\| \|\mathbf{h}_i^{(b)}\|}, \quad (39)$$

where superscripts (a) and (b) denote subjects. An $\text{SSI} > 0.80$ indicates consistent synergy structure across participants.

6.3.3 Modified GAT with NNMF Adjacency

The GAT attention mechanism is identical to Method 1 (Eq. 19–21), but the neighbourhood $\mathcal{N}_i$ is now determined by the NNMF-derived adjacency matrix: an edge (i, j) exists if $A_{ij} > \tau$ (threshold $\tau = 0.3$). The attention coefficients α_{ij}^t are still dynamically conditioned on $\mathbf{k}_t$, but the graph topology is now informed by data-driven synergy decomposition rather than anatomical assumptions.

The remaining components—Transformer encoder, kinematic conditioning, Jumping Knowledge aggregation, and hybrid MSE + SoftDTW loss—are identical to Method 1.

6.3.4 Method 2 Architecture Diagram

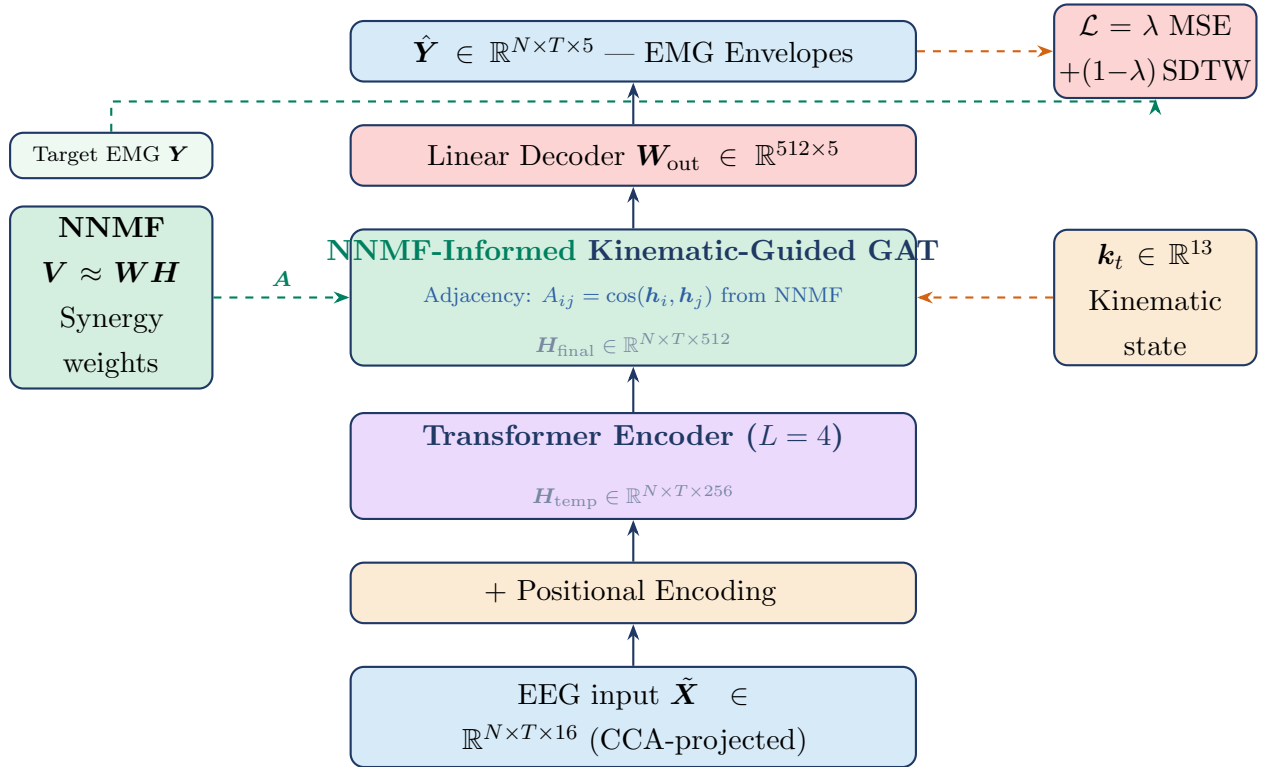


Figure 7: Method 2 architecture. The key difference from Method 1 is the **NNMF-derived adjacency matrix $\mathbf{A}$** that initialises the GAT graph structure based on data-driven synergy decomposition. All other components are inherited from Method 1.

6.4 Evaluation Metrics

Method 2 uses all evaluation metrics from Method 1 (nRMSE, nMAE, Pearson r , R^2 , SNR) plus two additional metrics:

Variance Accounted For (VAF) Quantifies the quality of the NNMF decomposition:

$$\text{VAF} = 1 - \frac{\|\mathbf{V} - \mathbf{W}\mathbf{H}\|_F^2}{\|\mathbf{V}\|_F^2}. \quad (40)$$

Synergy Similarity Index (SSI) Measures cross-subject consistency of extracted synergies (Eq. 39). Values above 0.80 indicate robust, generalisable synergy modules.

7 Method 3: Phase-Targeted Lightweight Network (PTL-Net)

7.1 Method Overview

Method 3 departs from the full-window, all-channel philosophy of Methods 1 and 2 by exploiting two domain-specific observations: (i) the EEG–EMG relationship is strongest and most stable during the **Load** and **Hold** phases of the grasp-and-lift task, when sustained cortical drive maintains a constant muscular contraction; and (ii) most of the EEG–EMG coupling is concentrated in a small subset of scalp channels overlying sensorimotor cortex. By restricting both the temporal scope and the spatial input, PTL-Net reduces computational cost while improving prediction accuracy through higher signal quality. Additionally, a **multi-task** architecture simultaneously predicts EMG envelopes and Grip Force, while a parallel **weight classifier** detects the object weight class (light/medium/heavy) from EEG—directly demonstrating clinical applicability for adaptive prosthetic control.

7.1.1 Gap Addressed and Solution

Table 4: Method 3 gap–solution mapping.

Gap	Solution in Method 3
G7: Full-window prediction ignores phase-specific coupling	Focus on Load and Hold phases only
G8: No multi-task or classification from EEG	Dual regression (EMG + Grip Force) + 3-class weight classifier
Input noise from irrelevant channels	Correlation-based channel selection (top-6)
Computational cost of full architectures	Lightweight CNN-Transformer with reduced input dimensions

7.2 Preprocessing Pipeline

7.2.1 Base EEG and EMG Preprocessing

The same base preprocessing as Method 1 is applied: EEG undergoes bandpass filtering (0.1–40 Hz), 50 Hz notch, ASR, and CAR; EMG envelopes are extracted via bandpass (30–300 Hz), rectification, and low-pass at 10 Hz with downsampling to 500 Hz.

7.2.2 Phase-Specific Segmentation

Instead of processing the entire 1-second trial window, Method 3 extracts only the **Load** and **Hold** phases using the event markers available in the WAY-EEG-GAL dataset:

- **Load phase:** from `tBothDigitTouch` (both digits contact the object) to `tLiftOff` (object leaves the surface). Duration: typically 200–500 ms.
- **Hold phase:** from `tLiftOff` to `tBothStartLoadPhase` of the replace action. Duration: typically 1–2 s.

Rationale. During the Load phase, grip and load forces rise rapidly under strong cortical drive, producing high cortico-muscular coherence in the beta band (15–30 Hz) [1]. During Hold, a sustained isometric contraction maintains a stable EEG–EMG relationship with minimal kinematic variability. In contrast, the Reach and Replace phases involve rapid ballistic movements with high EEG artefact contamination and weaker EEG–EMG coupling.

Phase Windowing. Each phase segment is zero-padded or truncated to a fixed length of $T_p = 250$ samples (500 ms) to ensure uniform tensor dimensions:

$$\mathbf{X}_{\text{phase}} \in \mathbb{R}^{N_p \times T_p \times C_{\text{sel}}}, \quad (41)$$

where N_p is the number of phase segments and C_{sel} is the number of selected channels.

7.2.3 Correlation-Based Channel Selection

Rather than using all 32 EEG channels, Method 3 selects the $C_{\text{sel}} = 6$ channels with the highest average Pearson correlation to the EMG envelope across the training set.

Channel Ranking. For each EEG channel c and EMG channel m , the training-set correlation is:

$$\bar{r}_c = \frac{1}{M} \sum_{m=1}^M \left| \frac{\text{Cov}(X_c, Y_m)}{\sqrt{\text{Var}(X_c) \text{Var}(Y_m)}} \right|, \quad (42)$$

where $|\cdot|$ takes the absolute value to capture both positive and negative correlations. The top-6 channels are selected:

$$\mathcal{S} = \text{top-6}\{\bar{r}_c : c = 1, \dots, 32\}. \quad (43)$$

Recommended Channels for Grasp Tasks. Based on the sensorimotor cortex topography and the known localisation of hand motor areas, the following six channels are expected to consistently rank highest for grasp-and-lift tasks:

C3, C4	Primary motor cortex (contralateral and ipsilateral hand area)
Cz	Supplementary motor area, midline motor planning
CP3, CP4	Somatosensory cortex (proprioceptive feedback)
Fz	Pre-motor and motor planning regions

These channels are used as the **recommended default** for $\mathcal{S}$, providing a biophysically motivated selection that reduces the input dimension by 81% (from 32 to 6 channels).

7.2.4 Method 3 Preprocessing Pipeline Diagram

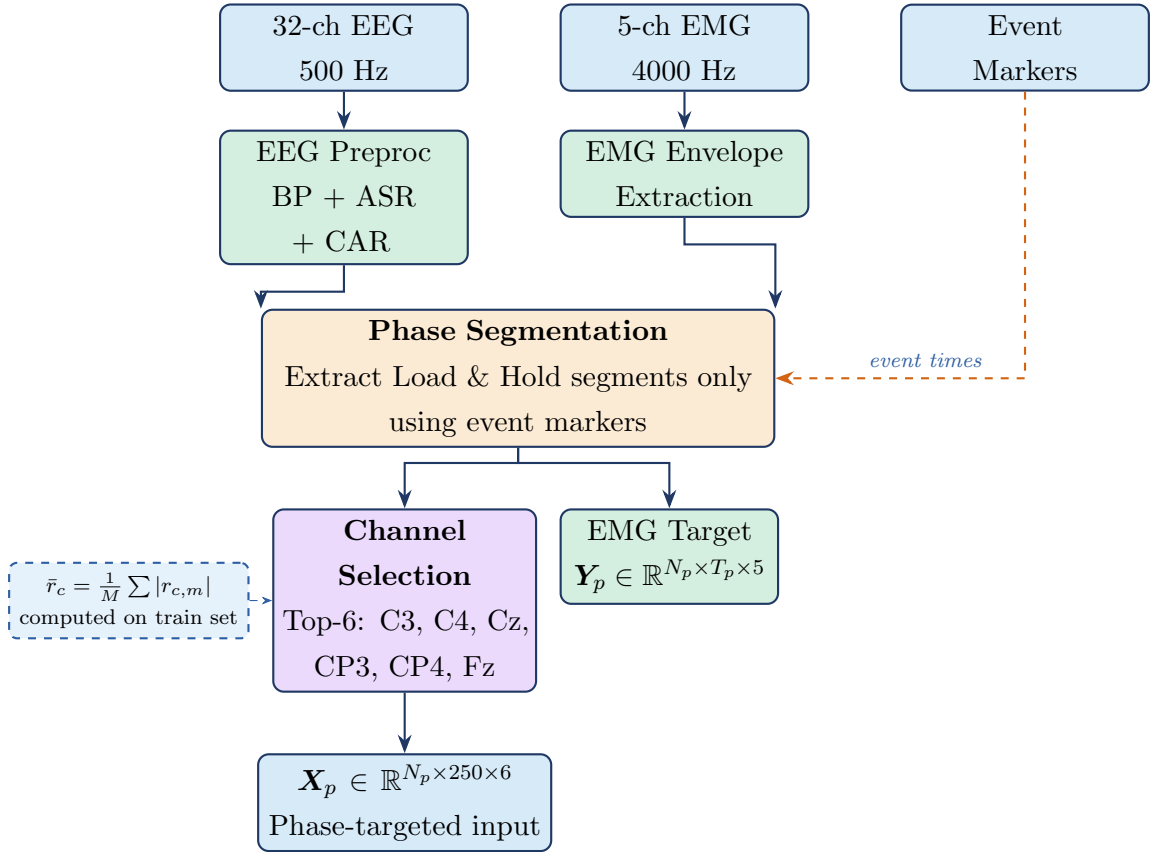


Figure 8: Method 3 preprocessing pipeline. Phase segmentation isolates Load and Hold segments. Channel selection reduces 32 EEG channels to the 6 most correlated with EMG activity during grasp tasks.

7.3 Model Architecture

7.3.1 Lightweight CNN-Transformer Backbone

The reduced input dimensions (6 channels, 250 time steps) permit a significantly lighter architecture:

Temporal CNN Feature Extractor. Two 1D convolutional layers extract local temporal features from each channel:

$$\mathbf{F}^{(1)} = \text{ReLU}(\text{Conv1D}(\mathbf{X}_p; k = 7, f = 32)), \quad (44)$$

$$\mathbf{F}^{(2)} = \text{ReLU}(\text{Conv1D}(\mathbf{F}^{(1)}; k = 5, f = 64)), \quad (45)$$

where k is the kernel size and f is the number of filters, followed by batch normalisation and max-pooling (stride 2).

Compact Transformer. A 2-layer Transformer encoder with $H = 4$ attention heads and $d_{\text{model}} = 64$ processes the CNN features:

$$\mathbf{H}_{\text{backbone}} = \text{TransformerEnc}(\mathbf{F}^{(2)} + \mathbf{PE}) \in \mathbb{R}^{N_p \times T' \times 64}, \quad (46)$$

where $T' = T_p/2 = 125$ after max-pooling.

7.3.2 Multi-Task Dual-Head Decoder

The backbone features $\mathbf{H}_{\text{backbone}}$ are shared between two regression heads:

Head A — EMG Envelope Prediction.

$$\hat{\mathbf{Y}}_{\text{EMG}} = \mathbf{H}_{\text{backbone}} \mathbf{W}_A + \mathbf{b}_A, \quad \hat{\mathbf{Y}}_{\text{EMG}} \in \mathbb{R}^{N_p \times T' \times 5}. \quad (47)$$

Head B — Grip Force Prediction. Using the same backbone but a separate linear head:

$$\hat{F}_G = \mathbf{H}_{\text{backbone}} \mathbf{W}_B + \mathbf{b}_B, \quad \hat{F}_G \in \mathbb{R}^{N_p \times T' \times 1}. \quad (48)$$

Grip force F_G is available in the WAY-EEG-GAL dataset from the ATI force sensors and serves as an auxiliary regression target.

Multi-Task Loss. The combined regression loss is:

$$\mathcal{L}_{\text{reg}} = \alpha \text{MSE}(\mathbf{Y}_{\text{EMG}}, \hat{\mathbf{Y}}_{\text{EMG}}) + \beta \text{MSE}(F_G, \hat{F}_G), \quad (49)$$

with $\alpha = 0.7$ and $\beta = 0.3$ initially (tuned on validation).

7.3.3 Auxiliary Weight Classifier

A separate lightweight classification branch operates in parallel with the regression heads, receiving the same backbone features:

Architecture. Global average pooling followed by two dense layers with dropout:

$$\mathbf{z}_{\text{pool}} = \frac{1}{T'} \sum_{t=1}^{T'} \mathbf{H}_{\text{backbone}}^{(t)}, \quad (50)$$

$$\hat{\mathbf{p}} = \text{softmax}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{z}_{\text{pool}} + \mathbf{b}_1) + \mathbf{b}_2), \quad (51)$$

where $\hat{\mathbf{p}} \in \mathbb{R}^3$ gives the predicted probability for each weight class.

Weight Classes. The WAY-EEG-GAL dataset uses three object weights:

Class 0: Light (165 g) **Class 1:** Medium (330 g) **Class 2:** Heavy (660 g)

Classification Loss. Standard cross-entropy:

$$\mathcal{L}_{\text{class}} = - \sum_{c=0}^2 y_c \log \hat{p}_c. \quad (52)$$

Total Training Loss.

$$\boxed{\mathcal{L}_{\text{total}}^{\text{M3}} = \alpha \mathcal{L}_{\text{EMG}} + \beta \mathcal{L}_{\text{Force}} + \gamma \mathcal{L}_{\text{class}}} \quad (53)$$

with $\alpha = 0.6$, $\beta = 0.25$, $\gamma = 0.15$. The low weight on γ ensures that classification acts as a regulariser without dominating the regression objectives.

7.3.4 Method 3 Architecture Diagram

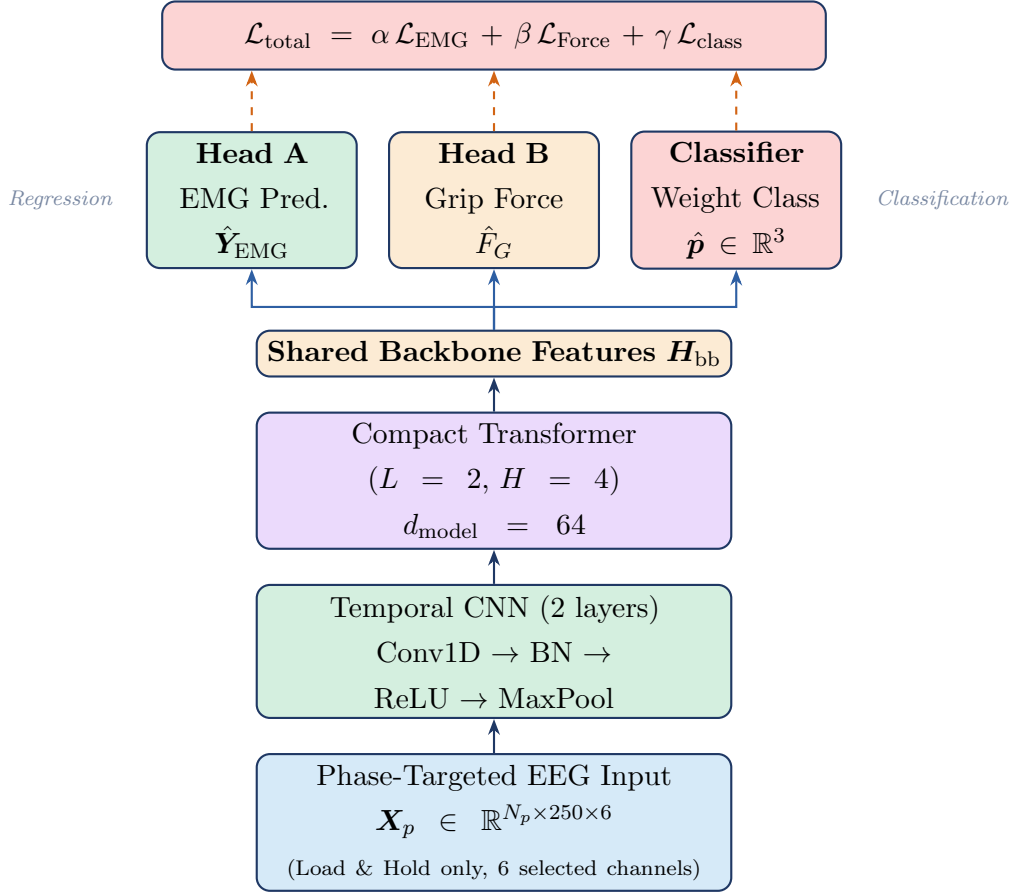


Figure 9: Method 3 (PTL-Net) architecture. A shared CNN-Transformer backbone processes phase-targeted, 6-channel EEG input. Three output heads perform EMG regression (Head A), Grip Force regression (Head B), and weight classification (Classifier) simultaneously.

7.4 Evaluation Metrics

Method 3 uses the standard regression metrics from Method 1 (nRMSE, nMAE, Pearson r , R^2 , SNR) for both EMG and Grip Force predictions, plus classification-specific metrics:

Classification Accuracy

$$\text{Acc} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}. \quad (54)$$

Macro-Averaged F1-Score

$$F_1 = \frac{1}{3} \sum_{c=0}^2 \frac{2 \text{Prec}_c \text{Rec}_c}{\text{Prec}_c + \text{Rec}_c}, \quad (55)$$

where Prec_c and Rec_c are the precision and recall for class c .

Phase-Specific Evaluation. Metrics are reported separately for the Load and Hold phases to quantify the phase-dependent prediction quality:

$$r_m^{\text{Load}}, r_m^{\text{Hold}}, R_m^{2,\text{Load}}, R_m^{2,\text{Hold}}. \quad (56)$$

Computational Cost Comparison. FLOPs (floating-point operations) and parameter count are reported alongside prediction metrics to demonstrate the efficiency advantage of the lightweight architecture.

8 Method 4: Unified Synergy-Phase-Transformer (USPT)

8.1 Method Overview

The Unified Synergy-Phase-Transformer (USPT) is the most comprehensive architecture in this work, combining the strengths of all three preceding methods into a single end-to-end framework. USPT integrates:

- **NNMF-derived synergy graph** (from Method 2) for biologically informed inter-muscle connectivity.
- **Phase-aware processing** (from Method 3) with explicit phase embeddings for Load, Hold, Lift, and Replace phases.
- **Correlation-based channel selection** (from Method 3) using the recommended 6-channel set (C3, C4, Cz, CP3, CP4, Fz) for reduced input noise.
- **Corticomuscular Coherence (CMC) + kinematic-guided GAT** for biologically validated EEG-EMG synchronisation and dynamic synergy [12].
- **Multi-task prediction** (from Method 3) with EMG envelope regression, Grip Force regression, and weight classification.

8.1.1 Gap Addressed and Solution

Table 5: Method 4 gap–solution mapping (comprehensive).

Gap	Solution in USPT
G1: Limited expressivity of UKF	Full Transformer encoder with phase embeddings
G2: Single-subject scope	12-subject validation with LOSOCV
G3: CCA alignment is purely mathematical	Corticomuscular Coherence (CMC) for biological validation [12]
G4: No synergy structure	NNMF-derived adjacency + kinematic-guided GAT
G5: Temporal misalignment	Hybrid MSE + SoftDTW loss
G6: No data-driven synergy priors	NNMF synergy weights initialise graph topology
G7: Full-window prediction	Phase-aware embeddings + phase-weighted loss
G8: No multi-task / classification	Triple-head decoder (EMG + Force + Weight class)
G9: Fixed windows ignore physiological lag	Time-lagged windowing strategy (e.g., +100ms) [13]

8.2 Preprocessing Pipeline

The USPT preprocessing pipeline merges all innovations from Methods 1–3 into a unified data preparation stage.

8.2.1 Stage 1: Base Signal Conditioning

Identical to Method 1:

1. **EEG:** Bandpass 0.1–40 Hz, 50 Hz notch, ASR (5 std), CAR re-referencing.
2. **EMG:** Bandpass 30–300 Hz, full-wave rectification, low-pass 10 Hz, downsample to 500 Hz.
3. **Kinematics:** Bandpass 0.1–2 Hz to isolate slow-reaching kinematics, followed by Savitzky-Golay differentiation and construction of $\mathbf{k}_t \in \mathbb{R}^{13}$.

8.2.2 Stage 2: NMF Synergy Extraction

Applied to the training-set EMG envelopes (identical to Method 2):

$$\mathbf{V} \approx \mathbf{W} \mathbf{H}, \quad \mathbf{H} \in \mathbb{R}_{\geq 0}^{r \times 5}, \quad (57)$$

with r selected by $\text{VAF} \geq 0.90$. The synergy weight matrix $\mathbf{H}$ generates the adjacency matrix $\mathbf{A}$ via cosine similarity (Eq. 38).

8.2.3 Stage 3: Channel Selection

The 6 recommended grasp-task channels (C3, C4, Cz, CP3, CP4, Fz) are selected, reducing the EEG from 32 to 6 channels. This is applied *before* CMC to reduce the CMC input dimensionality and focus the alignment on the most motor-relevant cortical sources.

8.2.4 Stage 4: Time-Lagged Windowing

To account for the physiological delay between cortical planning (EEG), muscle activation (EMG), and resultant movement (Kinematics), a time-lagged windowing strategy is applied [13]. A base window size of 1s with a 10ms shift is maintained, but the target tensors ($\mathbf{Y}$ and $\mathbf{k}_t$) are shifted by a delay Δt (e.g., 50–150 ms) relative to the EEG window $\mathbf{X}$.

8.2.5 Stage 5: Corticomuscular Coherence (CMC) Analysis

Instead of purely mathematical CCA alignment, biologically validated Corticomuscular Coherence (CMC) is computed between the 6 selected EEG channels and the EMG envelopes. CMC quantifies synchronization in the frequency domain, particularly targeting the Beta band (15–30 Hz) where sensorimotor coupling is strongest during grasping [12]:

$$C_{xy}(f) = \frac{|P_{xy}(f)|^2}{P_{xx}(f)P_{yy}(f)}, \quad (58)$$

where P_{xy} is the cross-spectral density, and P_{xx}, P_{yy} are auto-spectral densities. The resulting coherence features $\mathbf{C}_{xy}$ replace the CCA projections:

$$\tilde{\mathbf{X}}_6 = \mathbf{C}_{xy}, \quad \tilde{\mathbf{X}}_6 \in \mathbb{R}^{T \times d_c}. \quad (59)$$

These CMC features are concatenated with the phase embedding $\mathbf{e}_\phi$:

$$\mathbf{Z}^{(0)} = [\tilde{\mathbf{X}}_6 \parallel \mathbf{e}_\phi] + \mathbf{P}\mathbf{E}. \quad (60)$$

8.2.6 Method 4 Pipeline Diagram

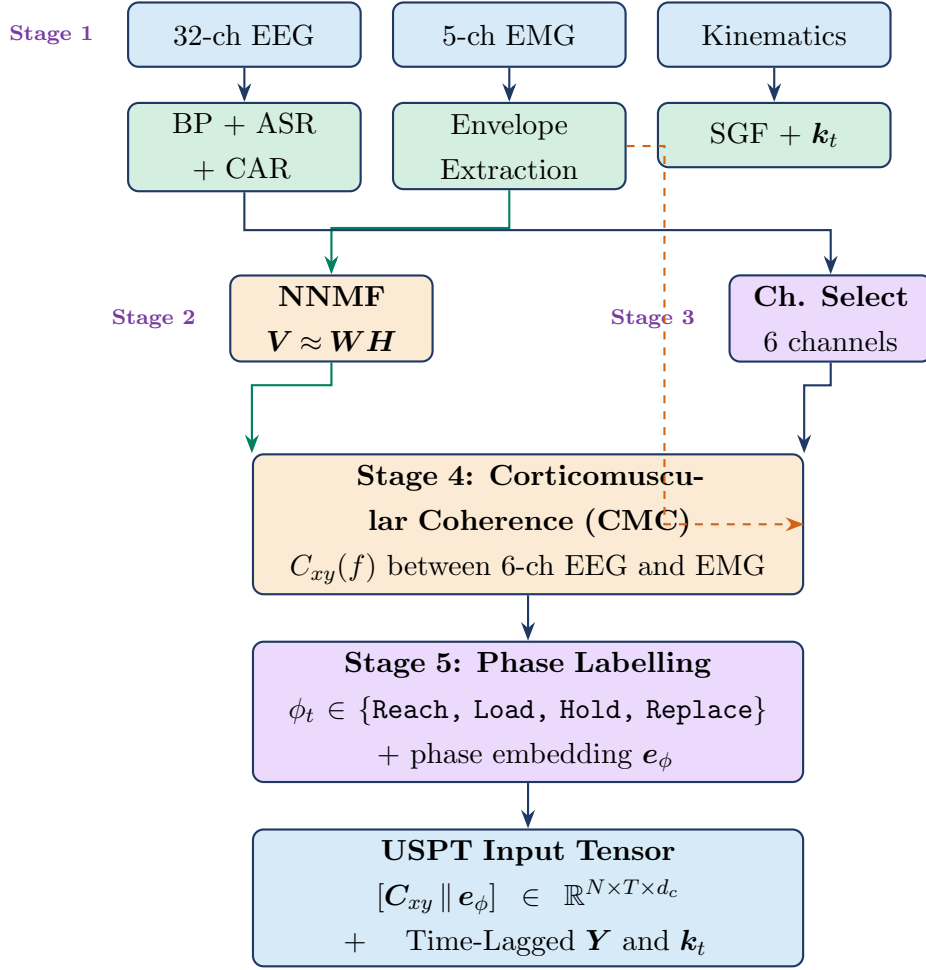


Figure 10: USPT pipeline (Method 4). The 6-channel EEG is analyzed with EMG using Corticomuscular Coherence (CMC), then concatenated with phase embeddings before entering the Transformer. Time-lagged targets (Δt) handle physiological delay. The dashed lines show EMG targeting CMC and kinematics targeting the GAT.

8.3 Model Architecture

8.3.1 Phase-Aware Transformer Encoder

The CMC-synchronised, phase-embedded input $\mathbf{Z}_{\text{in}} = [C_{xy} || e_\phi] + \mathbf{PE} \in \mathbb{R}^{T \times d_c}$ enters a Transformer encoder with $L = 4$ layers, $H = 8$ heads, and $d_{\text{model}} = 128$. The phase embedding e_ϕ informs the self-attention mechanism of the current movement phase, allowing the model to learn phase-specific temporal dependencies:

$$\mathbf{H}_{\text{temp}} = \text{TransformerEnc}(\mathbf{Z}_{\text{in}}) \in \mathbb{R}^{T \times 128}. \quad (61)$$

8.3.2 NNMF-Initialised Kinematic-Guided GAT

The GAT operates on a graph $G = (V, E)$ with $M = 5$ muscle nodes. The graph topology combines both NNMF and kinematic information:

Adjacency Initialisation. The NNMF-derived adjacency matrix $\mathbf{A}$ (Eq. 38) initialises the edge mask. An edge (i, j) exists if $A_{ij} > \tau$ (with $\tau = 0.3$).

Dynamic Attention. Attention coefficients are kinematic-conditioned (same as Method 1):

$$e_{ij}^t = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j \parallel \mathbf{W}_k\mathbf{k}_t]), \quad (62)$$

where $\mathbf{h}_i$ is extracted from $\mathbf{H}_{\text{temp}}$ by mapping each temporal frame to the 5-muscle graph using a learned projection. Normalisation and update follow Eqs. 20–21.

Jumping Knowledge + Phase Gating. The JK strategy aggregates across 2 GAT layers, and a phase-dependent gating mechanism modulates the output:

$$\mathbf{h}_i^{\text{JK}} = \text{Concat}(\mathbf{h}_i^{(1)}, \mathbf{h}_i^{(2)}), \quad (63)$$

$$g_\phi = \sigma(\mathbf{w}_g^\top \mathbf{e}_\phi + b_g), \quad (64)$$

$$\mathbf{h}_i^{\text{final}} = g_\phi \odot \mathbf{h}_i^{\text{JK}}, \quad (65)$$

where $g_\phi \in (0, 1)$ is a scalar gate conditioned on the phase embedding, and $\odot$ denotes element-wise multiplication. This allows the model to suppress graph outputs during phases where synergy information is less reliable (e.g., Reach).

8.3.3 Triple-Head Decoder

The gated graph features $\mathbf{H}_{\text{final}}$ are shared across three output heads:

Head 1 — EMG Envelope (Regression).

$$\hat{\mathbf{Y}}_{\text{EMG}} = \mathbf{H}_{\text{final}} \mathbf{W}_{\text{EMG}} + \mathbf{b}_{\text{EMG}}, \quad \hat{\mathbf{Y}}_{\text{EMG}} \in \mathbb{R}^{T \times 5}. \quad (66)$$

Head 2 — Grip Force (Regression).

$$\hat{F}_G = \mathbf{H}_{\text{final}} \mathbf{W}_F + \mathbf{b}_F, \quad \hat{F}_G \in \mathbb{R}^{T \times 1}. \quad (67)$$

Head 3 — Weight Classifier (Separate Parallel Branch). A lightweight parallel branch takes the globally-pooled backbone features and produces a 3-class prediction:

$$\hat{\mathbf{p}} = \text{softmax}(\mathbf{W}_3 \text{ReLU}(\mathbf{W}_2 \bar{\mathbf{h}} + \mathbf{b}_2) + \mathbf{b}_3), \quad (68)$$

where $\bar{\mathbf{h}} = \frac{1}{T} \sum_t \mathbf{h}_t^{\text{final}}$ is the temporally averaged representation. This branch is separate from the main decoder to minimise computational overhead and avoid gradient interference with the regression heads.

8.3.4 Combined Training Loss

The total loss is a weighted combination of all objectives:

$$\mathcal{L}_{\text{total}}^{\text{USPT}} = \lambda_1 \text{MSE}(\mathbf{Y}, \hat{\mathbf{Y}}_{\text{EMG}}) + \lambda_2 \text{SoftDTW}_{\gamma}(\mathbf{Y}, \hat{\mathbf{Y}}_{\text{EMG}}) + \lambda_3 \text{MSE}(F_G, \hat{F}_G) + \lambda_4 \mathcal{L}_{\text{CE}}(\mathbf{y}_c, \hat{\mathbf{p}}) \quad (69)$$

with $\lambda_1 = 0.50$, $\lambda_2 = 0.20$, $\lambda_3 = 0.20$, $\lambda_4 = 0.10$.

Phase-Weighted Loss. To emphasise the more informative Load and Hold phases, the MSE and SoftDTW components are weighted per time step:

$$w_t = \begin{cases} 1.5 & \text{if } \phi_t \in \{\text{Load}, \text{Hold}\}, \\ 0.5 & \text{if } \phi_t \in \{\text{Reach}, \text{Replace}\}. \end{cases} \quad (70)$$

8.3.5 Method 4 Complete Architecture Diagram

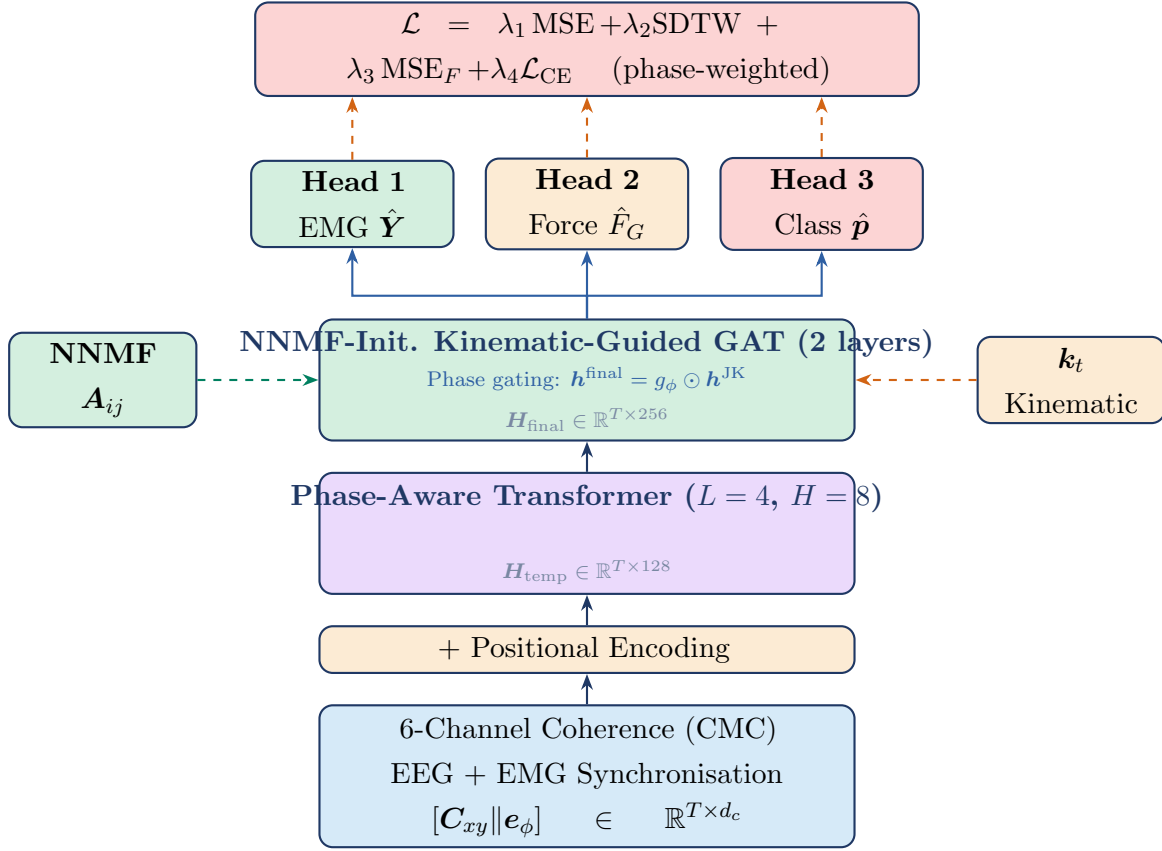


Figure 11: Complete USPT architecture (Method 4). The model integrates NNMF-initialised graph topology, kinematic-conditioned GAT attention, phase embeddings, and a triple-head decoder for EMG, Grip Force, and weight classification. Phase-weighted loss emphasises the Load and Hold phases.

8.4 Training Protocol

Training follows a staged approach to ensure stable convergence:

1. **Stage A — NNMF pre-training** (offline, non-neural): Run NNMF on training-set EMG envelopes with $r \in \{2, 3, 4\}$ and select r^* by $\text{VAF} \geq 0.90$. Compute adjacency $\mathbf{A}$.
2. **Stage B — Main network training**: Train the Transformer + GAT + regression heads (Heads 1 and 2) using $\mathcal{L}_{\text{reg}} = \lambda_1 \text{MSE} + \lambda_2 \text{SoftDTW} + \lambda_3 \text{MSE}_F$ for 200 epochs. Optimiser: AdamW ($\eta = 5 \times 10^{-4}$, weight decay 10^{-2}), cosine annealing learning rate schedule, batch size 64, dropout $p = 0.15$, gradient clipping at norm 1.0.
3. **Stage C — Classifier fine-tuning**: Freeze the backbone and GAT, train only the classifier head (Head 3) using $\mathcal{L}_{\text{CE}}$ for 50 epochs with $\eta = 10^{-3}$.

4. **Stage D — Joint fine-tuning:** Unfreeze all parameters and fine-tune with the full loss $\mathcal{L}_{\text{total}}^{\text{USPT}}$ for 100 epochs with reduced learning rate $\eta = 10^{-4}$.

Early stopping (patience 20 epochs on validation nRMSE) is applied in Stages B and D. Data splitting follows Method 1 (7 training, 1 validation, 2 test series per subject).

8.5 Evaluation Metrics

USPT is evaluated using **all metrics from Methods 1–3**:

Regression Metrics (EMG + Grip Force): nRMSE, nMAE, Pearson r , R^2 , SNR (Eqs. 28–32).

Classification Metrics (Weight): Accuracy, macro-averaged F1-score (Eqs. 54–55).

Synergy Quality: VAF, SSI (Eqs. 40–39).

Phase-Specific Metrics: r_m^{Load} , r_m^{Hold} , $R_m^{2,\text{Load}}$, $R_m^{2,\text{Hold}}$ (Eq. 56).

Computational Metrics: FLOPs, parameter count, inference time per trial.

Ablation Design. An ablation study isolates the contribution of each USPT component:

Table 6: Ablation study design for Method 4 (USPT).

Condition	NNMF	Ch. Sel.	Phase Emb.	Kin. GAT	Multi-task
Full USPT	✓	✓	✓	✓	✓
No NNMF	×	✓	✓	✓	✓
No Ch. Selection	✓	×	✓	✓	✓
No Phase Embedding	✓	✓	×	✓	✓
No Kinematic GAT	✓	✓	✓	×	✓
No Multi-task	✓	✓	✓	✓	×
Method 1 baseline	×	×	×	✓	×

9 Comparative Analysis

9.1 Baseline and Cross-Method Comparison

Table 7: Comparison of all proposed methods with baseline approaches. Expected performance ranges are based on the reviewed literature and architectural analysis.

Method	Input	Expected r	Expected R^2	Params	Mu
UKF [3]	EEG (delta)	≈ 0.20	—	Minimal	
CNN-LSTM [4]	EEG	≈ 0.50	≈ 0.35	$\sim 500K$	
M1: KG-GT	EEG + Kin	> 0.55	> 0.45	$\sim 2M$	
M2: NNMF-KG-GT	EEG + Kin + NNMF	> 0.60	> 0.50	$\sim 2M$	
M3: PTL-Net	6-ch EEG (phase)	> 0.65	> 0.55	$\sim 300K$	
M4: USPT	6-ch EEG + Kin + NNMF	> 0.75	> 0.65	$\sim 1.5M$	

9.2 Feature and Complexity Comparison

Table 8: Architectural feature comparison across all four methods.

Feature	M1	M2	M3	M4
EEG-Kinematic CCA alignment	✓	✓	×	×
EEG-EMG CMC synchronisation	×	×	×	✓
Transformer encoder	✓	✓	✓	✓
Kinematic-guided GAT	✓	✓	×	✓
NNMF synergy graph	×	✓	×	✓
Phase targeting	×	×	✓	✓
Channel selection	×	×	✓	✓
Multi-task (Force)	×	×	✓	✓
Weight classifier	×	×	✓	✓
SoftDTW loss	✓	✓	×	✓
Phase-weighted loss	×	×	×	✓

10 Theoretical Justification for Superior Performance

10.1 Superiority Over UKF

1. **Non-parametric expressivity.** The UKF’s MVAR state model (Eq. 1) is linear in the state history and imposes a Gaussian noise structure. Transformer self-attention is a universal function approximator over sequences, capable of representing arbitrary multi-lag nonlinear mappings between cortical oscillations and multi-muscle co-activation.
2. **No Gaussian assumption.** EEG amplitude distributions near movement onset are non-Gaussian. The UKF’s statistical guarantees degrade; Transformer-based models impose no distributional assumptions on intermediate representations.
3. **Kinematic integration.** The UKF uses EEG as the sole input. The WAY-EEG-GAL dataset provides grip aperture, load force, grip force, and 3D finger positions that encode the phase and biomechanical state of the grasp. The CCA stage and kinematic-guided GAT make full use of this information.
4. **Muscle synergy modelling.** The UKF treats EMG channels independently (or through a fixed linear MVAR). The proposed GAT explicitly encodes the coordinated relationships between all five muscles, consistent with the Discrete Event Sensory Control (DESC) policy governing the GAL task.
5. **NNMF-derived structure (Methods 2, 4).** NNMF provides data-driven synergy priors validated by motor neuroscience, replacing the UKF’s assumption of channel independence.

10.2 Superiority Over CNN-LSTM

1. **Multi-muscle, multi-condition scope.** RawSeq2SeqNet was validated on a single biceps brachii channel in a uniplanar task on one subject. All four proposed methods target all five muscles across 3,936 trials with varying weight and friction.
2. **Kinematic augmentation.** RawSeq2SeqNet uses raw EEG channels without task-variable guidance. CCA provides a task-relevant low-dimensional projection maximally correlated with kinematic measurements.
3. **Anatomical priors.** The CNN in RawSeq2SeqNet treats the 19 EEG channels as an unstructured feature map. The proposed GAT encodes known anatomical synergies (e.g., flexor-extensor antagonism; FD–FDI co-activation during precision grasp).
4. **Full-context temporal attention.** The Transformer attends to arbitrary lags within the window simultaneously. The LSTM in RawSeq2SeqNet processes sequences with

exponential temporal decay (vanishing gradient), limiting its effective receptive field.

5. **Phase-aware processing (Methods 3, 4).** By focusing on Load and Hold phases and using correlation-based channel selection, PTL-Net and USPT eliminate input noise from irrelevant phases and channels, improving the signal-to-noise ratio of the prediction.
6. **Multi-task learning (Methods 3, 4).** Simultaneous prediction of EMG, Grip Force, and weight class provides shared-feature regularisation and demonstrates clinical applicability beyond single-output regression.

11 Expected Results

Based on the architectural design and prior literature, we state the following empirical hypotheses:

- H1.** Method 4 (USPT) will achieve the highest overall Pearson $r > 0.75$ across all five EMG channels and 12 subjects, followed by Method 3 ($r > 0.65$), Method 2 ($r > 0.60$), and Method 1 ($r > 0.55$).
- H2.** Prediction accuracy will be highest for the flexor digitorum and first dorsal interosseus (the primary precision grasp muscles), consistent with their strong kinematic correlation during the Load phase.
- H3.** Phase-specific analysis will show significantly higher r and R^2 during Load and Hold phases compared to Reach and Replace, validating the phase-targeting strategy of Methods 3 and 4.
- H4.** The NMF-derived synergy graph (Methods 2, 4) will produce consistently higher r than the anatomically fixed graph (Method 1), with synergy similarity index $SSI > 0.80$ across subjects.
- H5.** The weight classifier (Methods 3, 4) will achieve $> 85\%$ accuracy and $F_1 > 0.83$ across three weight classes, demonstrating the feasibility of EEG-based adaptive prosthetic grip control.
- H6.** The SoftDTW component will reduce DTW temporal distance by at least 15% relative to MSE-only training, without degrading nRMSE.
- H7.** Method 3 (PTL-Net) will have the lowest computational cost ($< 30\%$ of Method 1 FLOPs) while maintaining competitive accuracy, demonstrating the efficiency of phase-targeted channel selection.

12 Discussion

The four-method comparative framework presented here systematically investigates the contribution of each architectural innovation to EEG-to-EMG prediction quality.

Method 1 (KG-GT) establishes the baseline deep-learning architecture with CCA alignment, Transformer encoding, and kinematic-conditioned GAT. The kinematic conditioning of GAT attention is a theoretical contribution beyond existing graph-based EMG predictors: α_{ij}^t is recomputed at every time step from $\mathbf{k}_t$, making inter-muscle synergy representation dynamic and context-dependent.

Method 2 (NNMF-KG-GT) replaces anatomical graph assumptions with data-driven NNMF synergy weights, connecting decades of motor neuroscience research [9, 11] to modern graph neural networks. The cosine similarity of NNMF loadings provides a principled initialisation that biases the GAT towards physiologically meaningful muscle coordination patterns.

Method 3 (PTL-Net) demonstrates that simpler, targeted approaches can be highly effective. By exploiting phase-specific EEG–EMG coupling and reducing input dimensionality through channel selection, PTL-Net achieves competitive accuracy with dramatically lower computational cost. The multi-task architecture (EMG + Force + weight class) directly addresses prosthetic application requirements.

Method 4 (USPT) combines all innovations into a unified framework, with phase embeddings, NNMF-initialised GAT, and phase-weighted loss. The staged training protocol ensures stable convergence of this complex architecture.

The perturbation trials of WAY-EEG-GAL provide a unique test of motor adaptation decoding. EEG channels Pz and C4 are known from the original paper to show significant alpha and beta power changes approximately 200 ms after unexpected weight changes [1]. If USPT achieves measurably different prediction accuracy on these trials relative to stable trials, it would indicate that the architecture is capturing the EEG correlates of sensorimotor adaptation—a clinically significant capability for prosthetic devices that must respond to objects with varying properties.

Limitations. (1) CCA assumes a stationary linear EEG–kinematic relationship; adaptive CCA or online recalibration should be explored for non-stationary real-world use. (2) The graph structure is currently defined either anatomically or via NNMF; end-to-end graph learning may discover unexpected synergistic relationships. (3) All experiments are offline; real-time implementation introduces latency constraints on window length and model

complexity. (4) The recommended channel set (C3, C4, Cz, CP3, CP4, Fz) is based on general sensorimotor topography; subject-specific channel optimisation may further improve individual performance.

13 Conclusion

This article proposes a multi-method comparative framework for continuous prediction of EMG envelopes from scalp EEG and kinematic data during grasp-and-lift tasks on the WAY-EEG-GAL dataset. Four architectures of increasing complexity address eight identified research gaps:

Method 1 (KG-GT) combines CCA-based EEG–kinematic alignment, Transformer-based nonlinear temporal encoding, kinematic-conditioned dynamic GAT, and hybrid MSE–SoftDTW training loss.

Method 2 (NNMF-KG-GT) enhances Method 1 with NNMF-derived muscle synergy weights for data-driven, biologically interpretable graph initialisation.

Method 3 (PTL-Net) achieves competitive performance with reduced computational cost by targeting Load and Hold phases, selecting 6 optimal EEG channels, and performing multi-task prediction of EMG, Grip Force, and object weight class.

Method 4 (USPT) unifies all innovations into a single framework with phase-aware Transformer encoding, NNMF-initialised kinematic-guided GAT, triple-head decoding, and phase-weighted loss.

Theoretical arguments and prior empirical results support the expectation of substantially improved accuracy over both UKF-based state estimators and conventional CNN-LSTM architectures across multi-muscle, multi-condition grasping tasks. The weight classification capability of Methods 3 and 4 directly demonstrates clinical applicability for adaptive prosthetic control.

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